

Addressing the Public Health Misinformation Challenge with Real-time Data Fusion

Anoud Bani Hani¹, Haleama Alsabbah², Munir Majdalawieh¹, and Nawel Bessadet³

¹ CTI, Zayed University, UAE,

² Public Health Department, College of Health Sciences, Abu Dhabi University, UAE.

³ Department of Public Health, Birmingham City University

nawel.bessadet@bcu.ac.uk

Abstract—Misinformation is at an all-time peak across the globe, it wouldn't be considered an exaggeration if we describe 'misinformation' as a billion-dollar industry where bad actors profit off the chaos generated by misinformation. The public health industry has been at the receiving end of this challenge for way too long, leading to a high mortality rate in the public health industry. The number of resources invested into misinformation has made this phenomenon complex through its adoption of new and sophisticated deception techniques. The impact of misinformation on democracy and human rights can be damning and have severe consequences, also counter-misinformation will prove to be counter-productive, further denting the integrity of democracy and human rights; the COVID-19 pandemic illustrates what the war of misinformation looks like, an unending wave of misinformation and impeding crack on democracy as we know it. On numerous levels, effective responses to tackle disinformation are required, some of which will include laws and regulations, civil actions, and corporate measures among other practices. The early months of 2020 undeniably altered humanity's lifestyle in several ways, some of which humans could not have imagined. Of all the changes brought upon humanity by the 2020 COVID-19 pandemic, the fusion of aggregated data using technological solutions is paramount. The adoption of AI in several fields was fast-tracked by the occurrence which is the COVID-19 pandemic; AI was already in the works, but the pandemic accelerated its wide-scale adoption by compressing digital transformation that would have taken two years into a few months. Here we review the misinformation challenge encountered by the public health sector and identify major gaps in research, we also propose a solution that leverages the IT techniques in AI, deep learning, and semantic technologies. The place of IT techniques in addressing the public health misinformation challenge with real-time data fusion will be explored in this paper.

Index Terms—Misinformation, Disinformation, Public Health Data, Artificial Intelligence, Semantic Technology, Deep Learning

I. INTRODUCTION

The diversity and huge volume of data provided by the World Wide Web are remarkable; audios, videos, and documents are some of the formats in which information exists on the internet. The publication of news and information online through unconventional means like articles and audio-visuals makes human expertise a requirement for detection and classification; existing computational techniques can be utilized in detecting unusual behaviours in content by separating malicious text in online content from fact-based content [1]. One of these techniques is natural language processing (NLP). There are other techniques, they are however designed to

analyse the propagation of falsified information when compared to real information [2]; extensively, this approach dives into how fake news and contents are propagated differently on networks when compared to an article with intact integrity. The response received by an article gets separated at a certain level to distinguish its validity. Other advanced methods can be employed to discover how social spaces react to an article/content while exploring other textual features to identify the nature of the subject content. COVID-19 is now seen by many as the first pandemic to be distributed across our world of hyper-connectivity, proving to be an occurrence capable of swiftly and rapidly impacting all the structures that uphold society. Irrespective of the numerous measures planted and implemented to curb the transmission of the COVID-19 virus according to the World Health Organization (WHO), hundreds of millions of cases were recorded with over two million recorded deaths across the globe. In the early periods of 2021 panic and concern levels regarding the healthcare system already peaked; the healthcare system was experiencing threatening demands for medical services with the available medical services being significantly low when compared to the demands. The availability of measures and techniques able to reduce the time frame and increase efficiency for medical procedures like diagnostics, enhanced monitoring, tracking, and the prediction of contagion stages and how it affects society while simulating the results of a containment strategy and a medical protocol will define a new era in the health sector. As we know, the execution of these medical tasks simultaneously for thousands and millions of patients is humanly impossible without help; this makes the adoption of IT technology through artificial intelligence a formidable solution to information challenges in the health sector.

The emergency existence of COVID-19 has made new models and prototypes available in the quest to trace infections [3] and predict how these new infections are diffused and how they will react to preventive measures [4] [28] [30]. AI has experienced advancements that were unimaginable in the last decade, but that is the beauty of science and technology, endless possibilities; because of all these advancements, all the challenges in the healthcare industry can be enthusiastic about a solution soon. The ocean of information made available through pervasive IT and the ever-increasing computational power has assisted AI in displaying its capability concerning all the problems in the healthcare industry. The web has

experienced in the last three decades, has embraced the use of social media platforms like Facebook and Twitter which has in turn provided a new avenue for the distribution of information in a way that humanity has never experienced in its history; the news outlets among other beneficiaries of the social media emergence gained a lot by making information available to its subscribers in real-time. We have witnessed the evolution of news media from newspapers to digital news like social media feeds, blogs, online news platforms, and other digital media; we also witnessed everything in-between like magazines and tabloids [5]. It has become easier for consumers to access the latest choice information as they are happening. Social media platforms have proven their power as a tool of political propaganda in the last two decades through platforms like Facebook and Twitter by forcing tailored and carefully crafted content down the minds of platform users; these platforms are characterized by the liberty they provide users when it comes to discussing matters important to the society such as education, democracy, and health and education. This level of user liberty also provides an opening for malicious activities such as propaganda-driven news and conversations; some media agencies are well equipped to start sensitive and explosive conversations on social media platforms by spreading fake news to support their point of view. In other cases, these agencies and individuals create controversial opinions and manipulate mindsets through their fake news.

The rest of the paper is organized as follow, section 2 discusses the challenges of misinformation in public health and analyses relevant literature. Section 3 delves into the proposed data fusion approach to address these challenges followed by a conclusion.

II. THE PUBLIC HEALTH MISINFORMATION CHALLENGE

The definition of misinformation is somewhat complex but for the sake of this paper misinformation in science and health can be simply put as whatever information goes against the educated consensus of the scientific community concerning a phenomenon. This indicates that the nature of truth and falsehood is constantly changing due to the discovery of advanced methods and new evidence. Similarly, disinformation requires deliberate and well-planned effort in spreading misinformation for numerous gain purposes which may include power, money, and/or reputation. Given the public health sector's susceptibility to outright winners and losers, where the losers suffer great loss to their finance and reputation, malicious practices thrive through disinformation in the public health sector. Disinformation is seen as an arm of misinformation in this paper because disinformation comes with a sense of intention likened to the characteristic of the bad actors that spread the information rather than the information itself.

The last decade has seen an immense increase in fake news dissemination, a notable period being that of the 2016 US general elections [6]. This new culture of distribution of unverifiable articles online has upset so many landscapes of human society not limited to politics; health, science, and sports have also suffered greatly at the hand of disinformation [7]. In

the financial market, fake news propagated as rumor can have severe consequences that could eventually lead to the collapse of businesses and economies, and in some cases, the market could inherently come to a halt [8]. The type of information we consume significantly influences all of our daily decisions and our perception of society and the world at large is mainly shaped by all of the information our eyes and mind consumed; it is therefore important that we watch what information we consume so that we are not tricked to make choices that we normally would not make in life. There are numerous shreds of evidence to prove customers' absurd reaction to fake news that wasn't discovered early [Soll, 2016]. The spread of the coronavirus is a good representation of this case, where fake news was being spread all over the world through the internet concerning the origin and other characteristic features of the virus [9]. The discovery of more fake content online created a more unpleasant situation in society; the identification of fake content online is tasking. Thanks to technological advancements, several computational methods have been developed to help in identifying fake content based on their textual content [10]. The requirement for human expertise in identifying fake articles and websites proves to be a major challenge for these techniques.

The behaviour and mode of learning concerning ailments, risks, and preventive measures are not lacking in the literature and educational resources [11]. Sources of information are numerous, providing us with options as to where we collect our information and knowledge about the well-being of individuals; family, friends, healthcare professionals, television, pharmaceuticals, and educational resources are some of the places we get our information from. However, the adoption of technology for healthcare practices has dictated online services as the go-to source for information as it cuts out the stress and fuss of other information collection mediums stated earlier. The year 2013 is a good example in history since more than 70% of adults living in the US turned to the World-Wide-Web for health-related information [12]. Regardless of a few individuals like those in the higher age or low income and education bracket being unable to utilize the internet for health information generation [13], the democratization of medicine by the internet is not in doubt. The internet can sometimes be mistaken for a single-layered platform, but nothing can be further from the truth as the internet contains a pyramid of platforms and strata of functionalities. We now delve into some of the numerous misinformation paths online.

Some users seeking health information are accustomed to cutting off the search engine and going directly to the domain site to find health-related information; a user visiting a site like Wikipedia or The Daily Time online to read up on certain health subjects instead of typing that health subject into a search engine like Google or Bing is known as taking the 'Direct to online sources' path. The quality of direct sources varies significantly from established platforms such as Centre for Disease Control and Prevention (CDC) online to online blogs by less experienced writers. In addition, search engines on the internet are the subject of the online misinformation path which is 'search'. Google and Bing are the most popular examples of search engines. The early years of Google were designed to discover pages on the internet that contained whatever words were the subject of a search after which a list of top sites

containing the subject is presented. In this new world of fast-paced technology advancements, search engines have grown so much that their complexity cannot be adequately quantified; some specialized functionalities like the identification of health information contribute to the complex nature of search engines. Of all the internet searches, 5% are health-related, and this number doubles in the days before an emergency department visit [14]. Individuals have testified to search helps in making better and more informed decisions regarding their health problems however, a major challenge to the discovery of useful health information online exists and involves the daunting task of deciding the symptoms or diagnosis to input into the search engine. An investigation carried out [15] concerning the search for health information online discovered individuals with incorrect knowledge mostly try to find more information and data on irrelevant online platforms to further support their initially incorrect knowledge and hypothesis; this is known as confirmation bias, a case where individuals intentionally make evidence supporting their prior belief the object of their attention. Confirmation bias is very dangerous given the infinite pool of information available online; evidence can therefore be discovered to support almost any hypothesis specifically in the health field [16]. This ocean of available information that can easily be accessed makes the separation of fiction from facts as well as the interpretation of findings very difficult and humanly draining.

A. The Spread of Misinformation

Concerning misinformation and disinformation, there are many bad actors responsible, sources that profit significantly from the mishaps that emerge from the dissemination of misinformation and disinformation some of which include politicians, news media, and gossip columns [17]. Unlike contagion models which are now more efficient in putting meaning to the spread of diseases, the process of completely understanding the misinformation pandemic is only at the embryo phase and we are yet to get into the deep waters. Over 3 million individuals were discovered by [2] to have spread 126,000 rumors on a social media platform called 'Twitter'; false information was discovered to diffuse further and faster in a significant manner, reaching a broader audience than true information. Similar outcomes have been discovered through studies that focused on health misinformation. The spread of information through a person-to-person medium, known as large-scale virality is also notably a rare occurrence; an investigation done by [18] on the structure of content distribution on social media platforms like Twitter and the possibility of virality being the mode of distribution compared to the broadcast method; a broadcast is simply put as a means of information dissemination where a large number of individuals receive the same information from the same source; the popularity of this information is dependent on the influence of the source. The popularity of disseminated information on social media platforms was also discovered to be primarily predicted by the largest broadcast, with the relativity of uncommon occurrence being attributed to viral cascades. Small-sized arts on the internet attributed with virality are also noticed for the broad media coverage. These informative data are an indication of the need for bodies with large social media presence to accredit the validity of health

information being shared; this responsibility must not be taken lightly. Also, individuals with a high number of followers on social media platforms must be encouraged and incentivized to distribute correct information, as this is a fruitful plan in the war against misinformation.

Numerous and diverse parties spread misinformation online for different purposes. Some of these parties could include huge corporations looking to shape the perspective of the public towards their product and others could be media agencies that help further the campaigns of political establishments by exposing propaganda to a larger audience through media bots. Understanding the source of misinformation at a human level individual is very important as it helps us know how to proceed with addressing the public health misinformation challenge in real time; most of the information available concerning misinformation sources was made available through studies of political behaviour and news. Another misinformation concern is the existence of echo chambers where individuals are provided with worldview reinforcements through information diets and the exacerbation of extremism; regardless of the growing consensus that the fear of political echo chambers might have been blown out of proportion, it will be unwise to turn a blind eye to the impact of echo chambers on information validity particularly in the public health sector which is impacted through societal perception.

B. Evaluating Online Public Health Information Integrity

Healthcare professionals globally have witnessed the shift of patients from being just passive receivers of health advice to taking up a more active role in the consumption of health information; patients now have easy access to tools on the internet that helps evaluate health information, giving people the liberty to search the web for more information concerning health related issues and not solely depending on the information provided by healthcare professionals. This freedom of information provided by the internet comes at great cost to the people who need the health information and to the public health sector at large; since laypeople do not have the expertise of healthcare professionals, their susceptibility to being manipulated by bad information provided by the research they do on their ailment is very high. The impact of health information accessibility can only be determined by the overall quality and integrity of online health information, the ability of laypeople to draw the right inference from health issues, and the cost of individuals drawing the wrong inference about health issues. Several research institutes have dedicated time and resources to the investigation and study of online health information and its quality. A meta-analysis evaluating the quality of online information was done by [19]; this study grouped quality criteria based on accuracy and comprehensiveness which resulted in a conclusion that the internet has a serious quality problem. Over time, however,

education studies have identified an increase in the quality of information on the internet; regardless over half of on-line health information is still identified as unreliable and catastrophic. This nature of online health information is a solid indication of its unreliability; individuals must therefore be solely depended upon to critically evaluate and discern whatever

health information they are consuming.

Furthermore, finding and understanding health information through electronic media towards more informed health decisions is referred to as eHealth literacy. There are several eHealth literacy tools available today; the department of national assessment conducted a comprehensive estimate of health literacy in 2003 resulting in the report of basic health literacy levels being found in 36% of US adults, this low health literacy is estimated to drain the economy of up to \$238 billion annually (Understanding the Health Literacy of America, 2009). People with low health literacy levels are subjected to more life-threatening diseases, delayed health care, healthcare denial, high mortality rate, and more hospitalization [20]. Another challenge is that a large number of the general population, not just those with low literacy use websites with little or no credibility when searching for health information; the readability and understandability of low-quality information make it a great choice to individuals over high-quality information as these low-quality information sources are easier to engage and understand. The reduction of scientific quality in information has proven to significantly increase engagement, indicating the complexity of the scientific. Humans do not like when things are made too complex to understand and science is naturally complex to the layman; this also hints at scientific creators making their contents more interactive and easier to understand, for what is the necessity of information if the target audience cannot engage or understand it.

III. INTRODUCTION DATA FUSION FOR ADDRESSING PUBLIC HEALTH MISINFORMATION

Based on the systematic review carried out directed at the evaluation of the number of harm linked to the utilization of online based health information [21]; this review displayed a significantly low number of online health contents linked to actual harm. This resource and findings as well as other educational research concerning online health information point to two possibilities, one is that individuals are being unknowingly harmed by the health information they discover on the internet while the other possibility is that people forget the source of what health information they have gathered at any point in time, they can thereby not classify such information as harmful or inaccurate. The harm done to individuals is projected to go higher as people follow unconventional and inaccurate medical advice for several reasons such as popularity or the need for a fast solution to health issues. The public health misinformation challenge that is real-time data fusion is a prominent one and one to be very concerned about. Given the large volumes of information being updated daily in the health sector, it is impossible to manually document these updates; this leads to a deficit in information credibility. However, technology provides a solution to this industry-threatening challenge by leveraging IT techniques; we propose a solution that utilizes artificial intelligence, semantic technology, and deep learning. These IT domains will be further reviewed to solidify the position of our solution in the public health misinformation challenge with real-time data fusion. The overall framework for a data fusion technique leveraging technologies such as artificial intelligence, semantic standards and deep learning is represented in Figure 1.

Data ingestion and storage ingests diverse data types for aggregation and storage. Ingested data from various health information media includes streaming and batch data. They are processed during aggregation and stored in readiness for extraction of insights. With deep learning, stored data are pre-processed for classification based on deep learning models such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Graph Neural Networks (GNN). These include classification based on a corpus of sentences and their corresponding labels. This works by taking outputs from previous times as input to feed the network again, so the output consists of values present in the memory thereby increasing classification performance and providing better results. Furthermore, data classification using data inter-relations to infer labels is facilitated, with clustered data graph nodes using edge distances or weights. The classifications are transformed into RDF triples based on pre-defined ontological schemas or models. This provides a means of validating the RDF triples against the ontological models to identify semantic discrepancies, ambiguities and redundancies.

These are flagged consequently and further processing on such data is discontinued. Other triples deemed appropriate are added to the semantic knowledge graph. The knowledge graph allows knowledge acquisition and discovery, thereby facilitating enquiry and engagement systems to answer relevant questions, notification system to notify users about information updates, and recommendation system which recommends inferences or knowledge relevant to a specific solution to users. The enquiry and engagement systems navigate the graph using the entities in the question asked by the user along with the indicated relations. The notification system keeps users informed of updates happening in the Knowledge Graph. It can notify users about new data from stakeholders in the public health domain, especially when it meets a user's interests according to past interactions. The recommendation system works on collaborative filtering and content-based filtering. Collaborative filtering uses the knowledge some individuals interact with to recommend to other people who have similar interaction records, while content-based filtering assumes that users may be interested in knowledge close to their past interactions. The system uses the unified method for the process, involving the integration of the semantic representation of entities and relations, as well as the connectivity information. The knowledge graph's connective structure guides the refining of the entity representation to predict the preference of a user.

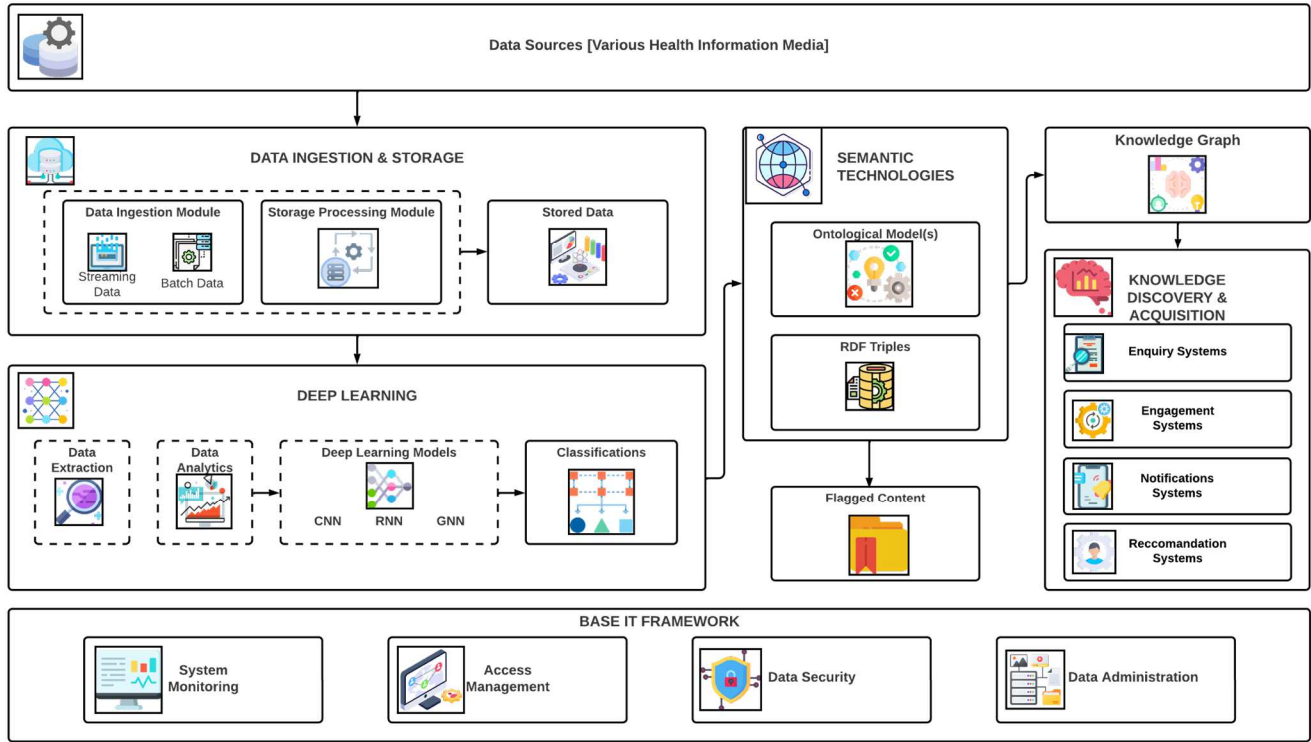


Fig. 1. Conceptual framework for a data fusion solution towards addressing public health misinformation

A. Artificial Intelligence

Artificial intelligence has been adopted across several industries and the public health sector is highly engaged with it. In the past, executing difficult tasks like emulating an individual or replicating a pattern used to demand lots of time and effort but AI makes such things easier to do with a lot of AI-supported programs. Implementing AI in healthcare activities projects potential for support with patient care and other administrative activities, helping healthcare practitioners with existing solutions and significantly reducing the time spent on overcoming challenges while increasing efficiency. Since AI is capable of learning and replicating human behaviour, healthcare practices are no different from AI. The adoption and application of AI in healthcare technologies vary between hospitals and healthcare facilities; the suggestion by some researchers concerning AI being more performing than humans in procedures like disease diagnosis, among others, we must be cautious so as not to eliminate the human factor in health care administration.

Furthermore, AI is still behind in its quest to completely replace humans in medicine. A large percentage of the public is still unsure of the role AI plays in healthcare and why it is so important for the future of medicine; Machine learning and diagnosis and treatment applications are some of the utilizations of AI in the healthcare industry. Machine learning stands as the basis for the most popular forms of AI in healthcare. Many approaches in AI generally utilize machine learning

since it is a vast technique. Precision medicine is where machine learning is already being used in healthcare; this process involves the accurate prediction of the best available treatment process for any ailment based on the patients' make-up and medical history, making the treatment framework an edge-cutting innovation for the public health sector. The last 50 years of healthcare advancement in technology have been directed toward the diagnosis and treatment of diseases, AI happens to have been a major contributor to this advancement in healthcare. The use of AI in healthcare for diagnosis and treatment plans independent of the mode of execution faces a major difficulty in collaborating with the clinical work process and EHR systems. The main challenge to AI adoption across all healthcare establishments is integration; this is a bigger problem compared to suggestions accuracy.

B. Semantic Technologies

The introduction of Semantic technology presented the solution to the information barriers on the web so that the comprehension and processing of information can be done by humans and not just machines. Overall, semantic technology is designed to encourage collaboration between humans and machines [22]. Semantic web technologies are tools capable of supporting system knowledge sharing while managing this relationship [23]. The whole idea surrounding semantic web is simply put as the creation of meaning and extensive knowledge for information; context is required, which semantic web technologies provide. Semantic web technology through all

its available tools and capabilities provides an environment for intelligent machine computing where all information, no matter how complex they are presented in a way that every member of the environment can understand; every piece of information in this environment is for human consumption and not the machine. Ontology, a concept broadly used in semantic technologies is a science rooted in the classification and cognition of concepts that originally belong to various fields; ontology is built on the philosophy of science. Ontology specialists describe ontology as an arm of philosophy, the science and structure of the types and object structures respectively, and all their characterizing attributes and how they relate to reality. Ontology essentially searches for entities and their classifications. Furthermore, standards ascertain consistency in information exchange while allowing collaboration and seamless interoperability between systems in semantic web technologies despite the provision of conceptual basis by ontologies to foster information exchange. The Resource Description Framework (RDF) is the basic standard on which interoperability is built; the RDF standard is object and XML oriented, designed for concept description and documentation in the semantic web. The XML is characterized by its support for multiple languages since Unicode and URI are its structural foundation.

The replication of knowledge and the practice of sharing knowledge by semantic technology efficiently contribute to scientific research and advancement through the invention, idea creation, hypotheses testing from various perspectives, creation of new researchers, and significantly cutting the cost of gathering information; semantic technology is the major contributor to the creation of relationships between several data sources through data mapping [27]. Global presence is a very important projection; that is, for healthcare services to be easily accessible to everyone every time has become an important goal. Online-based platforms have taken ubiquity very seriously by making information available at anytime, anywhere in the world. Hence, the collaboration between semantic web technology and web-based platforms promises to yield unprecedented results in the dissemination of unadulterated information across the healthcare space. Ontologies will maintain their position of importance in the healthcare information sector. The challenge of interoperating information systems in the healthcare sphere cannot be talked down on. This challenge exists and tackling it will spell a breakthrough for the industry. The importance of interoperability is tied to the free movement of information in a complex environment. The Dark Age meant the healthcare establishment had the wealth of data but very little information because most of the medical records existed in hard physical copies and were archived in an enumerable amount of files. The computerization of medical records, however, makes these would-be files exist in electronic formats where the data are structured in well-organized formats, unstructured formats, and visual data, all of which are integrated into healthcare organizations [24]. The integration of data into healthcare organizations thereby provides an opportunity for health information to be shared between all health organizations. Through semantic technology and the use of ontologies, verified health data can be seamlessly shared between every healthcare establishment, and the public health sector can benefit greatly

from this. Seeing as we'll like to tackle the public health misinformation challenge head-on, this provides the public health and the general public at large with an opportunity to extensively eradicate health misinformation in our society. Healthcare facilities are filled with experienced health professionals, so the information coming from these places can mostly be trusted; if health information from all medical establishments is bad then the health information space is in real trouble. With semantic ontologies, information is gathered from all healthcare providers and the free flow of verified information is established, gradually getting rid of misinformation in the public health sector. Authors of [29] [31] proposed Semantic Knowledge Graph for Context-Based Fake News Detection on Social Media.

C. Deep Learning

Deep learning is one of the most adopted sections of machine learning in the healthcare industry. The neurons in the human brain are the basis of deep learning; millions of neurons exist in the human brain with a hundred thousand links between them. Deep learning is a machine learning method known as an artificial neural network. The input layer, hidden layer, and output layer make up the neural network; there are numerous neurons in each layer with the hidden layer having the highest number of layers. There are numerous ways in which deep learning is being adopted in the medical field today, some of which include selection criteria, genomics, and medical imaging. Of all these applications of deep learning in the healthcare industry, medical imaging has been prevalent over the last three decades; the advancement and growth of the medical field and devices in this have resulted in the generation of more medical images like computed tomography (CT) and magnetic resonance imaging among others [25].

In addition, the ability to effectively and accurately analyse medical imaging is strongly dependent on varying degrees of experience, knowledge, other intangible factors that only experts possess, and time, and when the medical data is analysed in most cases, the human factor which is biased finds a way to creep in and flaw the analysis. Deep learning through the help of algorithms can cut out the human bias factor in its analysis and to a certain degree help experts in automated analysis. An algorithm may not have the ability to carry out the same task when faced with more complex medical imaging challenges that's why it is important that we do not eliminate the human factor in the analysis of medical images.

IV. CONCLUSION

The challenge of public health with real-time data fusion exists and this has created a gap that leads to misinformation in the public health sector. Adopting artificial intelligence, semantic technology, and deep learning provides a path to health information credibility and structure; if all of these are properly implemented, misinformation will be significantly reduced in the health sector, and we will be a few steps closer

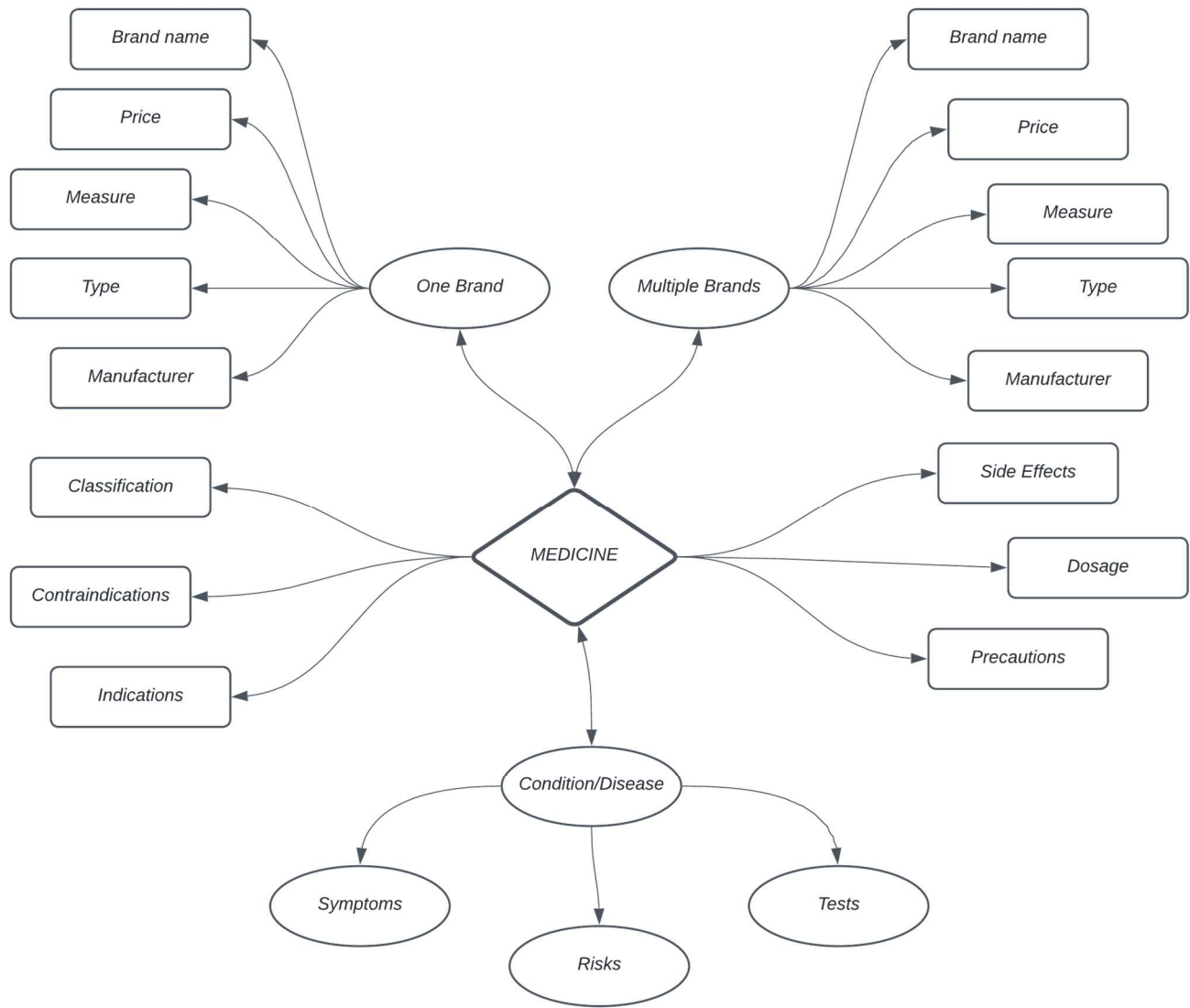


Fig. 2. High-level semantic representation for a sample topical public health domain, medicine

to completely eradicating societal health misinformation. We must embrace technological advancements in the healthcare industry and not distance ourselves from them because avoidance of new technology amounts to not taking the fight to bad actors and malicious individuals. Every available and formidable resource must be employed in this fight and technology has provided us with one.

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